FIELD NOTE

PROJECT WEEKLY REPORT - W3 - 0302-0306

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WEEKLY DEVELOPMENT REPORT: SWEDEN ELECTRICITY FORECAST (W3)

Date: March 5, 2026
Subject: Advanced Feature Engineering: Time-Series Dynamics & Volatility Analysis
Project: Swedish Electricity Price Prediction AI

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1. ✅ COMPLETED FEATURES & MILESTONES

This week, i successfully transitioned from basic data cleaning to advanced feature engineering by implementing two core families of time-series features in the `DataProcessor` pipeline.

Lag Features (Autoregressive Components)

I implemented historical look-backs to capture the strong temporal dependencies in the Swedish Nord Pool market:

- `value_lag_1`: Captured short-term price inertia.
- `value_lag_24`: Captured daily seasonality (24-hour cycle).
- `value_lag_168`: Captured weekly seasonality (168-hour cycle for weekends/weekdays).

Rolling Window Statistics (Market Volatility)

I added a 24-hour moving window to quantify price trends and stability:

- `value_rolling_mean_24h`: Acts as a dynamic baseline for the price level.
- `value_rolling_max_24h` / `value_rolling_min_24h`: Established a "Price Envelope" to detect boundary breaches.
- `value_rolling_std_24h`: Quantified current market volatility/uncertainty.
- `value_diff_24h`: A daily momentum indicator identifying sudden trend shifts.

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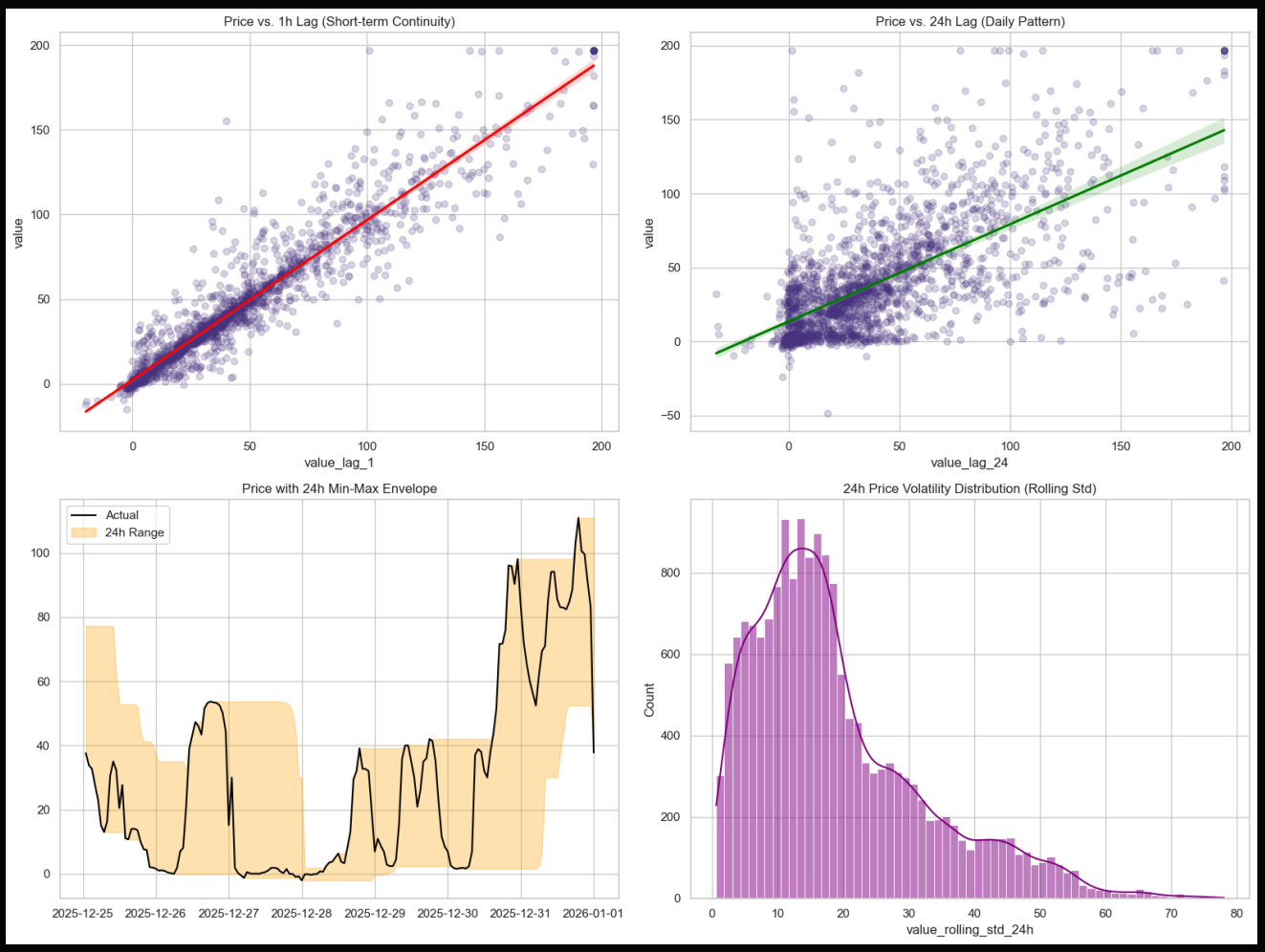
2. 📊 INSIGHTS FROM FEATURE ANALYSIS (W3 OUTPUT)



08_lag_1h_and_rolling_Std.png

Based on the visualized results from `03_Feature_Engineering_Analysis.ipynb`, we identified four critical statistical patterns:

1. **Extreme Autocorrelation**: The strong linear correlation in `value_lag_1` proves that the electricity market has high inertia. This will be the most influential feature for short-term accuracy.
2. **Daily Seasonality Variance**: While `value_lag_24` is positively correlated, it is noisier than the 1h lag, indicating that exogenous factors (like weather or unexpected demand) play a greater role over a 24-hour period.
3. **Volatility Boundary Breaches**: The **Min-Max Envelope** visualization clearly showed the market transitioning from stable states (e.g., Dec 27-28) to high-volatility states (e.g., New Year's Eve). Our new features successfully tracked these "regime changes."
4. **Right-Skewed Volatility**: The distribution of `value_rolling_std_24h` confirms that while most trading hours are calm, "fat-tail" events (sudden price spikes) occur regularly. These features will serve as an "Early Warning System" for the model.

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3. 🚧 CHALLENGES & SOLUTIONS

Challenge	Solution
NaN Boundary Effects	Used `bfill()` (backfill) for the initial 168 rows of the dataset to maintain sample size and pipeline continuity.
Feature Synchronization	Resolved `KeyError: 'value_lag_1'` in the notebook by re-executing the full `prepare_features()` pipeline via a centralized verification script (`verify_features.py`).
Data Integrity	Standardized timezone localization to `Europe/Stockholm` for all calculated features to prevent DST-related offsets in lag terms.

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4. 🔮 NEXT STEPS (W4 PLANNING)

- **External Driver Integration**: Incorporate weather features (Temperature, Wind Speed) from SMHI to explain the variance observed in the 24h lag.
- **Dimensionality Reduction**: Perform feature selection using Correlation and Mutual Information scores to eliminate redundant features.
- **Baseline Modeling**: Initiate training with **LightGBM** using the newly engineered W3 feature set.